CS1200 / CSCI E-120

Sorting algorithms and computational problems

Hesterberg–Vadhan textbook: Sections 2.5, 3.6 , 3.1 , 3.2 , Appendix A.3

A sorting exercise

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- Group 4: $((i, a), (2 + \frac{i}{2}, b), (-i, c), (1 + 3i, d), (1 - 3i, e), (2 + i, f))$

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- Time to solve a problem depends on our definition of 'elementary operations' (Group 3's experience).
- Some inputs can be sorted very easily (Group 2's input).

Computational problem

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- $\mathcal{I}$ is a (typically infinite) set of possible inputs (a.k.a. *instances*) x, and $\mathcal{O}$ is a (sometimes infinite) set of possible outputs y.
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  - $\mathbb{R} \to \mathbb{C}$
  - $f(x) =$ {All valid sortings of x} (if all keys are real), the empty set ϕ (otherwise)

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Note: *A fundamental point in the theory of algorithms is that we distinguish between computational problems and algorithms that solve them*

Measuring efficiency

- We first define a size parameter, a function $\mathrm{size} : \mathcal{I} \to \mathrm{R}^{\geq 0}$.
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The *(worst-case) running time for fixed-size inputs* of A :

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- Other notions of efficiency: how much space an algorithm needs; energy efficiency in the age of GenAI.

MergeSort

Solves the SORTING problem.

```
MergeSort(A):
Input  : An array A = ((K_0, V_0), ..., (K_{n-1}, V_{n-1})), where each K_i ∈ ℝ
Output : A valid sorting of A
0 if n ≤ 1 then return A;
1 else
2 |   i = ⌈n/2⌉
3 |   B = MergeSort(((K_0, V_0), ..., (K_{i-1}, V_{i-1})))
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MergeSort makes $O(n \log n)$ comparisons (next lecture).

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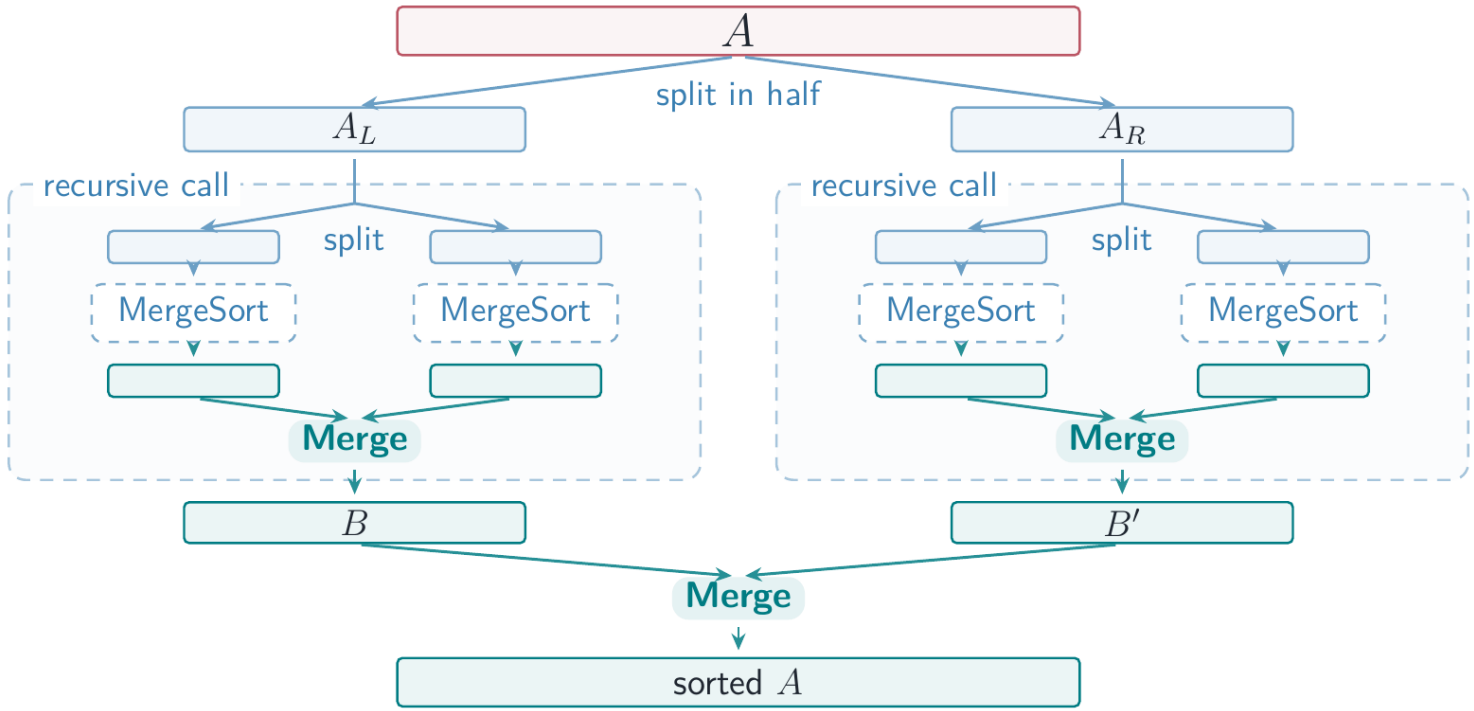
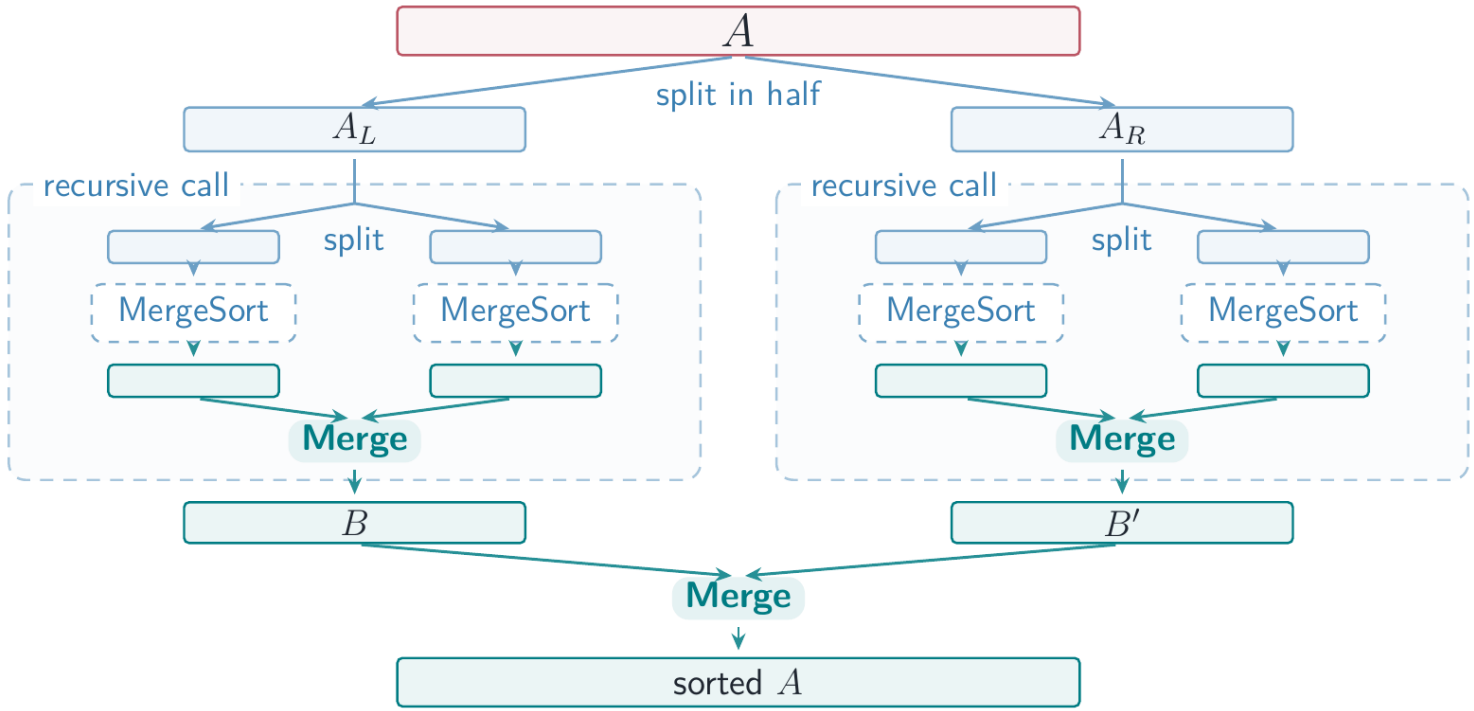

A
split in half
A_L
A_R
recursive call
split
MergeSort
MergeSort
Merge
recursive call
split
MergeSort
MergeSort
Merge
B
B'
Merge
sorted A

MergeSort



Proof of correctness: using strong induction (see Theorem 2.7 in Hesterberg-Vadhan).

Sorting on Finite Universe

Input: A universe size $U \in \mathbb{N}$ and an array A of key-value pairs $((K_0, V_0), \ldots, (K_{n-1}, V_{n-1}))$, where each key $K_i \in [U]$

Output: An array A' of key-value pairs $((K'_0, V'_0), \ldots, (K'_{n-1}, V'_{n-1}))$ that is a valid sorting of A.

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Singleton Bucket Sort (also called Pigeonhole sort): an algorithm to solve SORTINGONFINITEUNIVERSE in time $O(n + U)$.

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Stability: sorted output maintains the ordering of Values (if there is an ordering).

Singleton Bucket Sort: $U = 5$

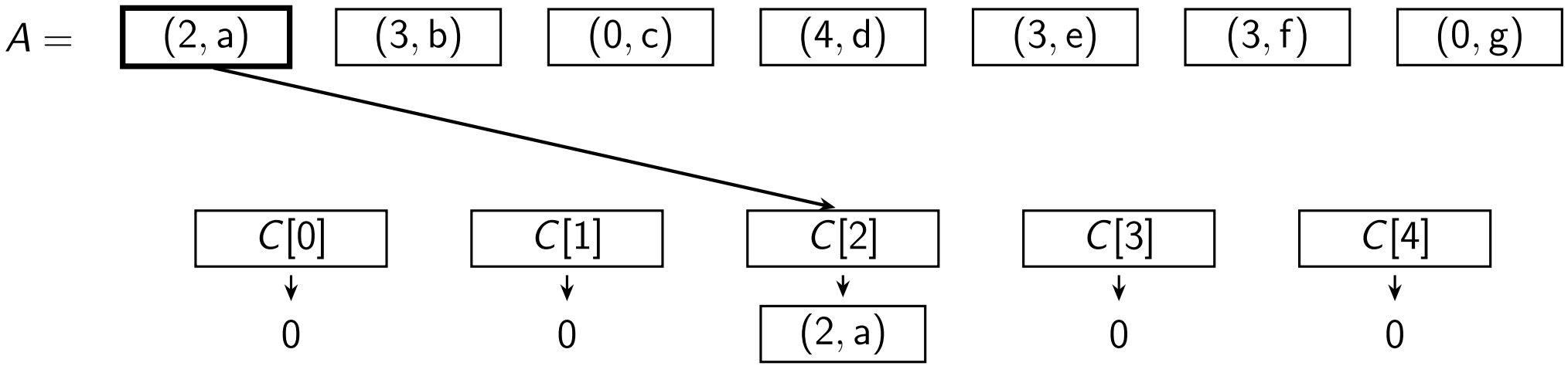
Initialize all five buckets to 0 (null).

$A =$ | $(2, \text{a})$ | $(3, \text{b})$ | $(0, \text{c})$ | $(4, \text{d})$ | $(3, \text{e})$ | $(3, \text{f})$ | $(0, \text{g})$ |

$C[0]$	$C[1]$	$C[2]$	$C[3]$	$C[4]$
↓	↓	↓	↓	↓
0	0	0	0	0

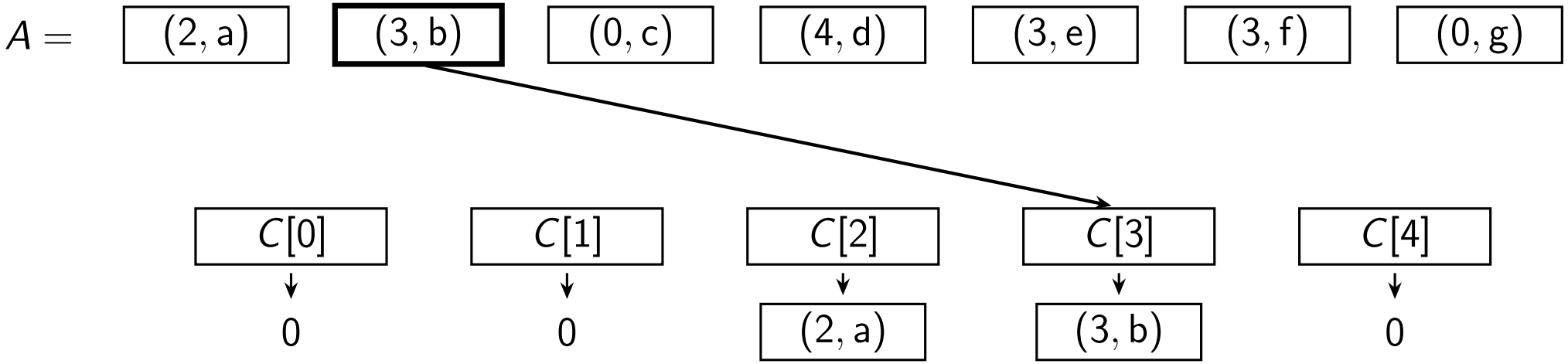
Singleton Bucket Sort: $U = 5$

Read A from left to right; insert at the head (top) of the list.



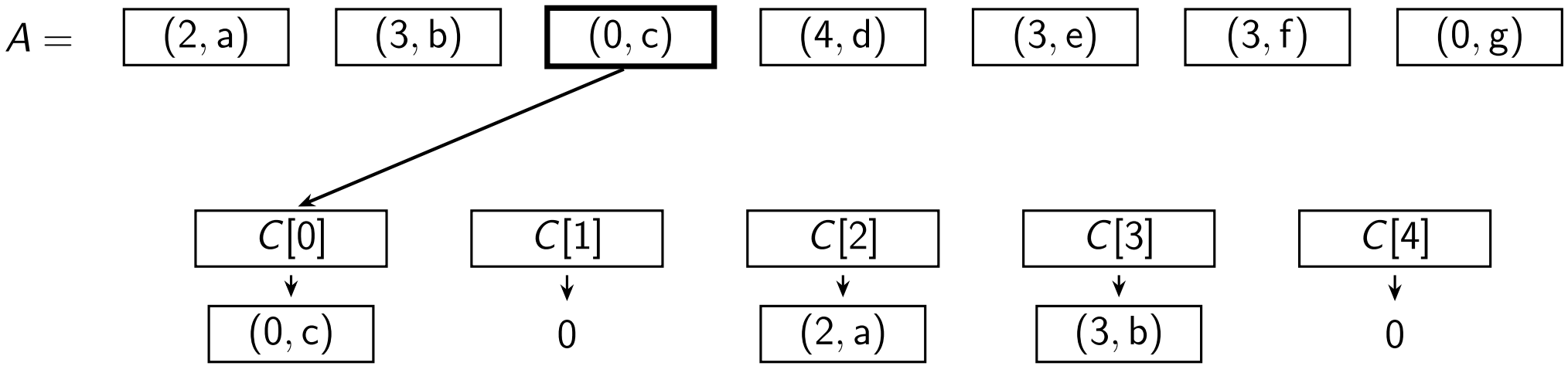
Singleton Bucket Sort: $U = 5$

Read A from left to right; insert at the head (top) of the list.



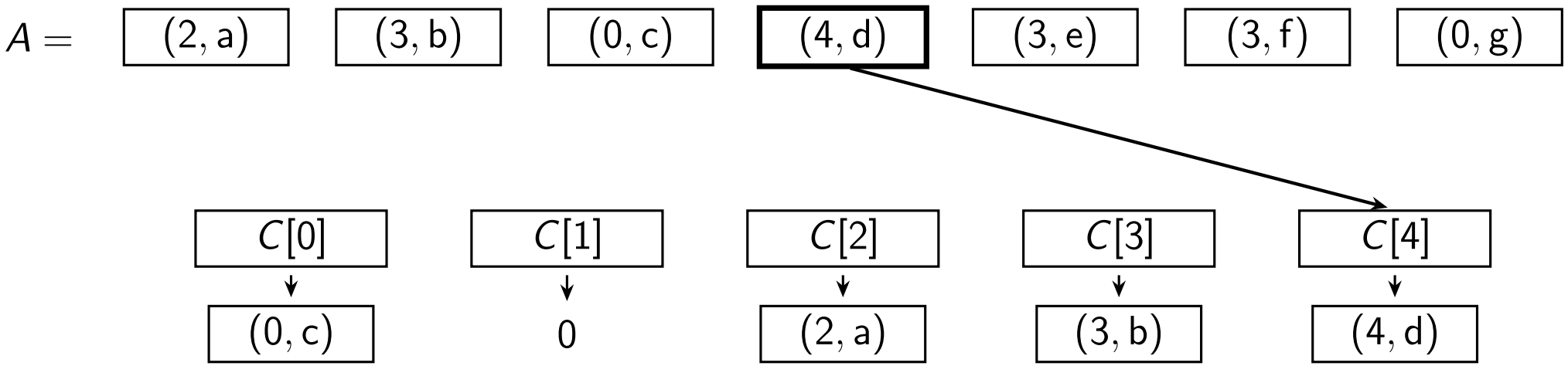
Singleton Bucket Sort: $U = 5$

Read A from left to right; insert at the head (top) of the list.



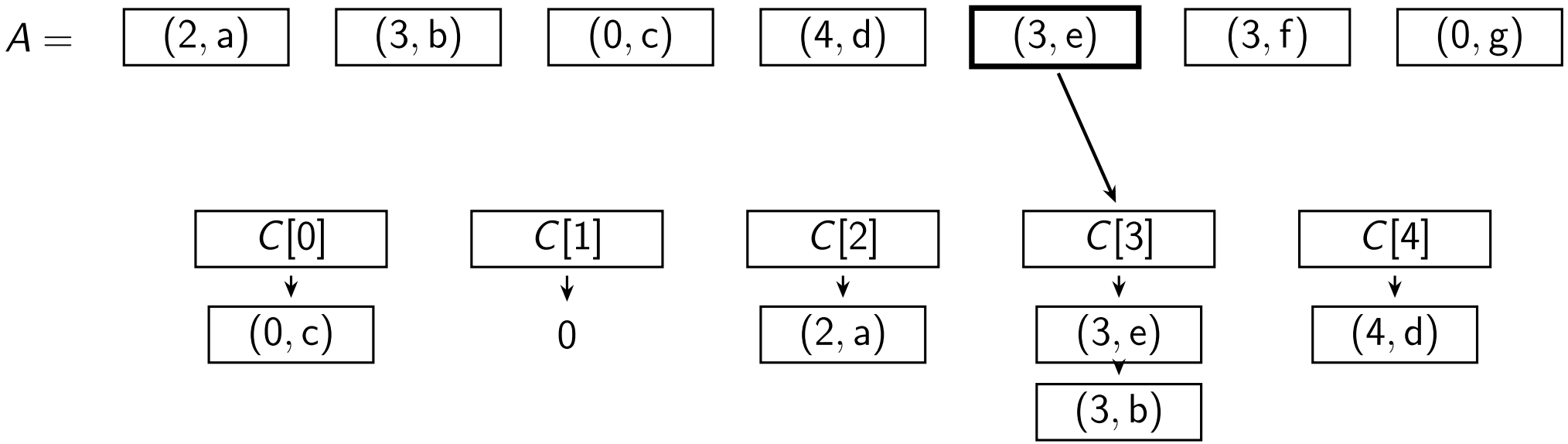
Singleton Bucket Sort: $U = 5$

Read A from left to right; insert at the head (top) of the list.



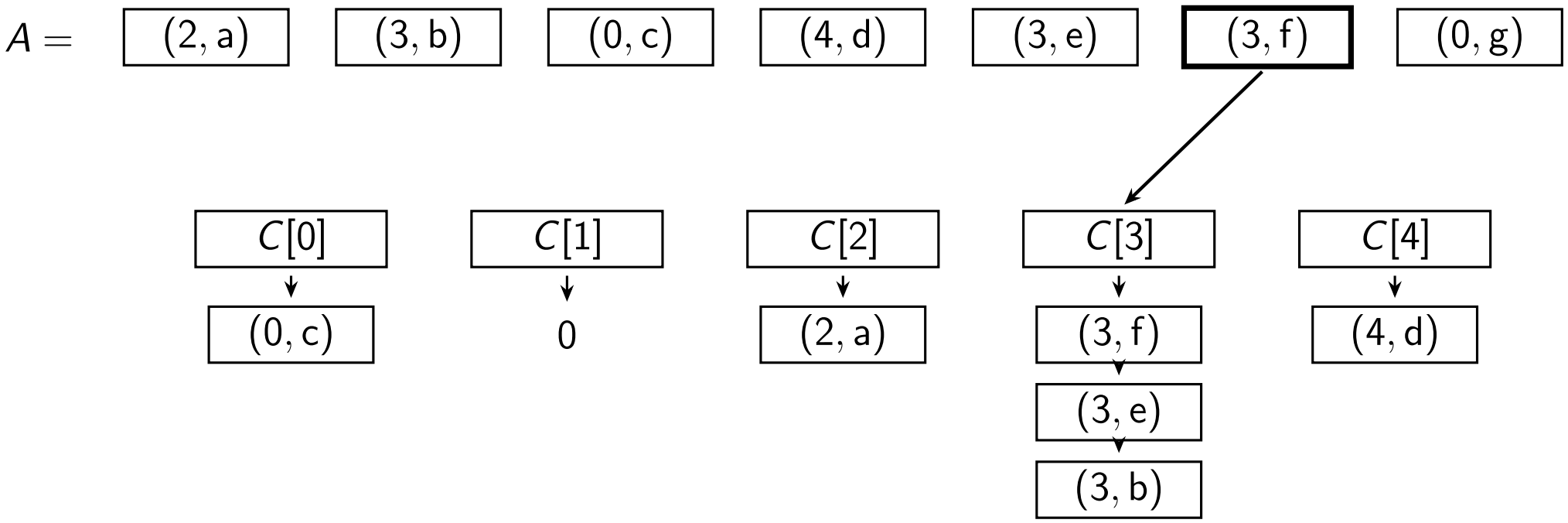
Singleton Bucket Sort: $U = 5$

Read A from left to right; insert at the head (top) of the list.



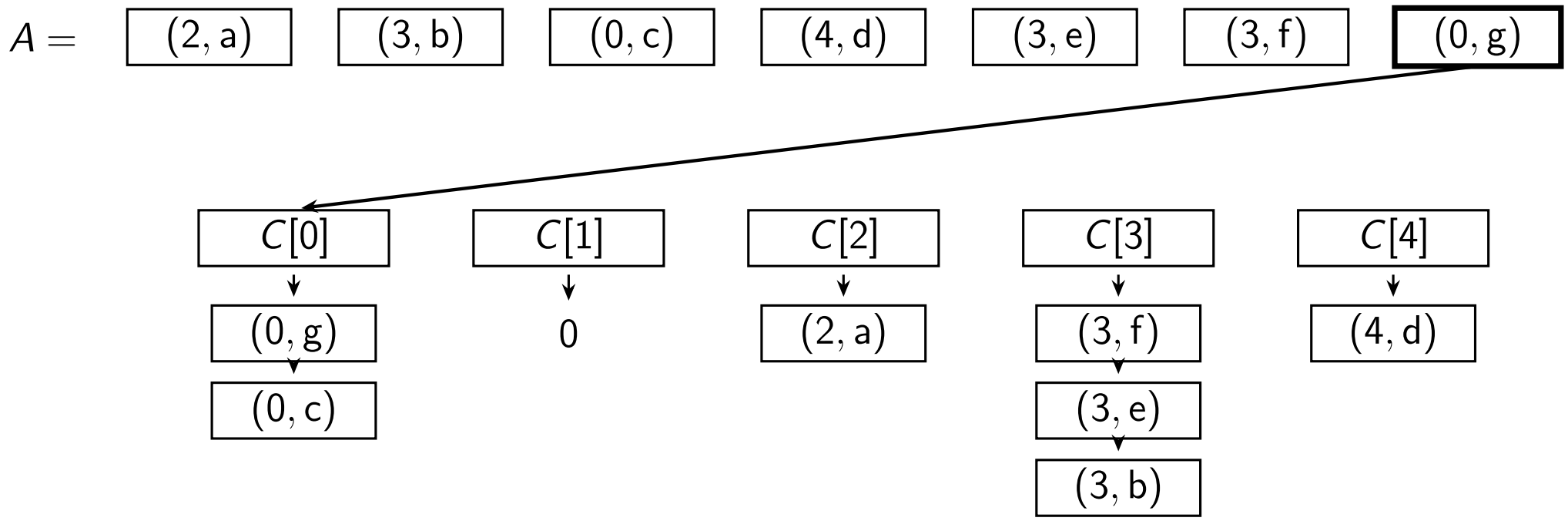
Singleton Bucket Sort: $U = 5$

Read A from left to right; insert at the head (top) of the list.



Singleton Bucket Sort: $U = 5$

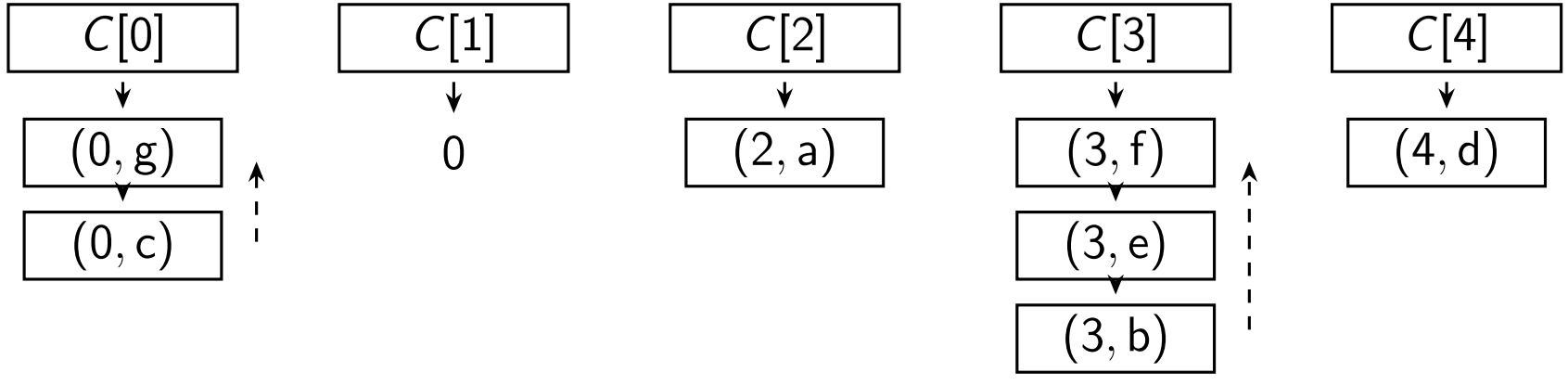
Read A from left to right; insert at the head (top) of the list.



Singleton Bucket Sort: $U = 5$

All records inserted. Lists are shown from head to tail, top to bottom.

$A =$ $(2, \mathrm{a})$ $(3, \mathrm{b})$ $(0, \mathrm{c})$ $(4, \mathrm{d})$ $(3, \mathrm{e})$ $(3, \mathrm{f})$ $(0, \mathrm{g})$



Visit $C[0], \ldots, C[4]$ in order; traverse each list from tail to head.

output: $(0, \mathrm{c})$ $(0, \mathrm{g})$ $(2, \mathrm{a})$ $(3, \mathrm{b})$ $(3, \mathrm{e})$ $(3, \mathrm{f})$ $(4, \mathrm{d})$